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Robust online multi-robot path planning for complex environments

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ABSTRACT

This work develops an adaptive enhanced framework to address slow convergence to high-quality trajectories, insufficient path smoothness, and elevated collision risk in multi-robot path planning. The approach is embedded in a multi-constraint, single-objective optimization model that incorporates kinematic limits, static and dynamic obstacle avoidance, and practical mission cost. Three complementary mechanisms are introduced: a velocity-alignment coordination to accelerate cooperative convergence of candidate solutions; an adaptive follow escape probability strategy that dynamically balances global exploration and local refinement to produce smoother and safer paths; and an adaptive random-perturbation mechanism to escape local optima and increase robustness in complex scenarios. Benchmark and simulation studies show that the proposed framework outperforms baseline and state-of-the-art comparison methods in terms of convergence accuracy, convergence speed, computational efficiency, path quality, and safety, providing a practical and robust solution for improving autonomy and operational performance in complex engineering multi-robot systems.

1. Introduction

Many fields, including intelligent manufacturing, warehouse logistics, environmental monitoring, and search and rescue operations, now widely use multi-robot systems [1–4]. In most of these applications, robots must work in a shared workspace. Therefore, it is critical for them to move safely and efficiently from a start point to a target. Robots must avoid collisions with static obstacles, dynamic threats, and other robots [5,6]. As a result, developing efficient path planning algorithms is crucial. These algorithms help minimize path length, reduce travel time, and ensure safety. This allows robots to finish tasks in dangerous or narrow areas [7,8]. Thus, solving this path planning problem is essential for increasing productivity, system safety, and seamless integration across many engineering and applied scientific fields.

Scholars have proposed various methods to solve these problems. Wang et al. proposed a method for random allocation of robot paths, using a fast-replanning algorithm and a potential field-based controller to avoid collisions [9]. Liu et al. combined UAV flight constraints with a sparse A* algorithm using a hierarchical optimization design [10]. Li et al. proposed a deep reinforcement learning method for UAV path planning in 3D environments [11]. In heuristic search, Lin et al. developed a method based on adaptive robot particle swarm optimization (RPSO) and gray wolf optimizer (GWO), which finds shorter paths with fewer iterations [12]. Similarly, Lin et al. proposed BPSO-RL, combining BPSO and Q-learning. Furthermore, this method is fast and supports dynamic scenes [13]. While these algorithms have their advantages, recent work shows that bio-inspired mechanisms can greatly improve robot mobility and adaptability in unstructured environments [14]. These studies suggest that imitating biological adaptation is key to better robustness in the real world. Unlike traditional deterministic methods A* or RRT which often rely on global map information and struggle with high-dimensional dynamic constraints in real-time, bio-inspired approaches offer superior flexibility and distributed coordination capabilities. Based on these biological inspirations, swarm intelligence algorithms are becoming a mainstream method in this field.

These algorithms are very suitable for multi-robot path planning in complex environments because they help robots learn from the environment and make effective strategies. Li et al. proposed an improved multi-step ant colony optimization (TDI-MSACO) to speed up convergence and shorten the path [15]. Wang et al. introduced an improved quantum particle swarm optimization (IQPSO) for underwater vehicles (AUVs), generating paths that are shorter and smoother than standard PSO [16]. To solve the global optimization issues of Particle Swarm Optimization (PSO), Yu and Huang introduced SDPSO and APSO algorithms. These methods improve search ability and plan better paths that fit the robot's movement limits [17,18]. Elmi et al. proposed a multi-

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objective GOA to minimize cost, distance, energy, and time, showing better path smoothness than PSO [19]. Also, Yang et al. proposed a hybrid chaotic game and grey wolf optimization (HCGO) for UAVs, achieving good results [20]. As optimization problems become more complex, scholars are introducing different strategies to improve algorithm performance. The Grasshopper Optimization Algorithm (GOA) is a relatively new method that has attracted attention due to its simple structure and good convergence [21]. However, traditional GOA has limitations when dealing with high-dimensional constrained problems. To fix this, Peng et al. added mathematical operators to refine the search [22], while Lei et al. developed an improved GOA using the Lévy flight mechanism (LFGOA) to increase randomness [23]. Hybrid strategies have also been proposed. Taher et al. proposed a Mixed GOA (MGOA) with mutation updates to expand the search space [24]. Zemmit et al. proposed WOA and GWO variants with adaptive step-size optimization to improve tracking accuracy and stability [25]. They also developed a GA-based parameter tuning method for control gains that reduces ripples and speeds up response [26], and Das et al. combined GOA with the *Dragonfly* Algorithm (DA) to improve performance [27].

Even though path planning algorithms have improved, dealing with complex environments is still hard. As environments get more crowded with moving obstacles and static ones, the chance of collision goes up. In such dense environments, it is difficult for traditional algorithms to quickly find the best path. This leads to slow response speeds and low efficiency. Also, the planned path is often not smooth enough, containing sharp turns or unnecessary shaking. These problems increase energy use and wear, limiting the robot's ability to move in narrow spaces. To solve these problems, this paper proposes an adaptive enhanced Grasshopper Optimization Algorithm, named AEM-GOA. Unlike traditional hybrid methods that rigidly combine two distinct metaheuristics, AEM-GOA adopts a modular adaptive architecture. The algorithm adds a velocity alignment mechanism to help the group work together better, allowing robots to quickly move toward the best solution. A follow-escape adaptive probability strategy is designed to balance global search and local refinement, producing smoother and safer paths. Also, an adaptive random perturbation mutation is introduced to help the algorithm escape local optima and ensure the robot finds the best path in complex environments.

The main contributions of this paper are as follows:

1 The architecture of GOA is reconstructed, and a multi-constraint, single-objective optimization model is established. This model integrates kinematic limits, dynamic obstacle avoidance, and static obstacle avoidance to achieve a unified solution for online multi-robot path planning.
2 Three enhancement mechanisms are introduced to improve performance. A velocity coordination mechanism is applied to accelerate collaborative convergence and ensure rapid robot response. A follow-escape adaptive probability strategy is designed to balance global and local search, generating smoother and safer paths. Additionally, an adaptive random perturbation mutation is used to help the robot escape local optima and avoid suboptimal paths.
3 The effectiveness of the AEM-GOA algorithm is fully verified through the CEC2019 standard test function set and multi-robot path planning simulations. Experimental results demonstrate that the algorithm exhibits strong robustness and optimization performance, making it highly effective for cooperative path planning tasks in complex environments.

2. Multi-robot path planning problem description and model

Online multi-robot path planning can be regarded as a constrained minimization problem. The goal is to reach the respective target point gradually at the minimum cost, under the premise that robots do not collide with static obstacles, dynamic obstacles, or companion robots. It is assumed that each robot in the robot system can be simplified as a circular rigid body with a radius r_i, and its position information at time t is recorded as $p_i^t = (x_i^t, y_i^t)$. The position change control quantity includes linear speed $v_i \in [v_{\min}, v_{\max}]$ and heading angle $\theta_i \in [\theta_{\min}, \theta_{\max}]$. Then the robot's position at the next time, t + 1, is updated to

$$p_i^{t+1} = p_i^t + v_i \begin{pmatrix} \cos(\theta_i) \\ \sin(\theta_i) \end{pmatrix} \tag{1}$$

The path planning problem in this paper is built on a well-recognized mathematical optimization framework, which uses the objective function proposed by Akay to dynamically constrain it [28].

The speed and angle are dynamically determined by the optimization algorithm in the iterative process. To comprehensively evaluate the quality and safety of multi-robot paths, a cost function with five evaluation criteria is constructed:

$$f1 = \left\| p_i^{t+1} - g_i \right\| = \sqrt{\left(x_i^{t+1} - x_i^g\right)^2 + \left(y_i^{t+1} - y_i^g\right)^2} \tag{2}$$

$$f2 = \left\| p_i^{t+1} - p_i^t \right\| = \sqrt{\left(x_i^{t+1} - x_i^t\right)^2 + \left(y_i^{t+1} - y_i^t\right)^2} \tag{3}$$

Among, $g_i = (x_i^g, y_i^g)$ are goal position coordinates of the robot *i*. For the *j* static obstacle, when the distance between the robot and it is less than the safe distance, a larger regularization term is constructed as a penalty constraint, otherwise 0:

$$f3 = \sum_{j=1}^{N_s} \begin{cases} \varepsilon & \left\| p_i^{t+1} - s_j \right\| < r_s \\ 0 & else \end{cases} \tag{4}$$

Among, s_j is the position of static obstacle,r_s is the safe distance,ε is the regularization,N_s is the number of perceived static obstacles. For the *j* dynamic obstacle, because of its initiative, when the distance between its position with the robot is less than the safe distance, the penalty function is constructed by the linear weighting of the distance difference:

$$f4 = \sum_{j=1}^{N_d} \max\left\{0, \varepsilon\left(a - \left(\left\| p_i^t - d_j^t \right\| - r_d\right)\right)\right\} \tag{5}$$

Among, a is the dynamic penalty regulatory factor, d_j^t is the position of dynamic obstacle, r_d is the safe distance,N_d is the number of perceived dynamic obstacles. Construct an index set of robots except this one, and design robot collision avoidance constraints for each robot $j \in \mathrm{I}_i$:

$$f5 = \sum_{j \in \mathrm{I}_i} \begin{cases} \varepsilon & \left\| p_i^{t+1} - p_i^t \right\| < r_r \\ 0 & else \end{cases} \tag{6}$$

When the safety distance between the two machines is less than r_r, a greater penalty is imposed. Therefore, the fitness function is the weighted sum of these fitness functions:

$$F = \min \omega_i (f1 + f2 + f3 + f4 + f5) \tag{7}$$

Among, ω_i is the weight that considers the mobile cost, various collision risks and task completion efficiency.

In each iteration, a local search space of speed range $[v_{\min}, v_{\max}]$ and heading range $[\theta_{\min}, \theta_{\max}]$ is constructed around each robot, and the designed AEM-GOA algorithm is used to search the control pair (v_i, θ_i) that minimizes F in this space. The robot moves in these feasible positions to reach the destination safely and effectively.

To effectively address this complex optimization model, a robust algorithmic framework must be established. Based on bio-inspired design principles and a performance evaluation integration mechanism [14], the following sections will elaborate on the proposed AEM-GOA algorithm, which aims to achieve an effective bridge between theoretical constraints and practical implementation.

3. Enhanced framework with adaptive cooperative

3.1. Principle of the grasshopper optimization algorithm

The GOA operates in two phases: a global exploration phase for identifying promising regions and a local exploitation phase for refining solutions. GOA adopts the social activities of exclusion and attraction according to the cluster idea, so that each grasshopper individual interacts to form a group movement [21].

In GOA, the individuals represent a solution, and the population's behavior is described by a mathematical model. The position X_i of each solution is calculated, and the formula is as follows:

$$X_i = S_i + G_i + A_i \tag{8}$$

Among, S_i represents interaction forces between agents, G_i represents influence of gravity, A_i represents guiding role of the wind, the S_i is as follows:

$$S_i = \sum_{j=1\, j\neq i}^{N} s(d_{ij}) * \widehat{d_{ij}} \tag{9}$$

Among, d_{ij} denotes the distance between the i th individual and the j th individual, $\widehat{d_{ij}}$ denotes the directional unit vector, s is a social force function, which is used to define the interaction between individuals:

$$s(r) = fe^{-\frac{r}{l}} - e^{-r} \tag{10}$$

Among, f is a parameter that controls the intensity of interaction between individuals, l is a parameter of the scope of action, and r expresses the mutual distance between individuals. Space between two grasshoppers can be divided into exclusion, comfort, and attraction zones by s. When the distance is greater than 10, the value returned by the function is close to zero. Therefore, the function cannot apply strength between grasshoppers with large distances and map the distance to [1,4]. G_i represents a centralized attraction component, the formula is as follows:

$$G_i = -ge_g \tag{11}$$

Among, g is the gravity constant, which controls the attraction strength, e_g is a unit vector pointing to ensure that everyone is uniformly attracted. A_i represents a directional bias vector on grasshoppers:

$$A_i = \mu e_w \tag{12}$$

Among, μ is drift constant e_w is wind vector. According to group force, gravity, and wind force, the updated formula of an individual can be obtained:

$$X_i = \sum_{j=1\, j\neq i}^{N} s\left(\left|x_j - x_i\right|\right) * \frac{x_j - x_i}{d_{ij}} - ge_g + \mu e_w \tag{13}$$

The number of people to control group is N. The formula (13) cannot be directly used to solve the optimization problem, mainly because the grasshoppers quickly reach the comfort zone, and the grasshopper population will not converge to the target position or the specified point. The adaptive target location progressively leads the individual to the optimal solution, helping to tackle the optimization challenge and preventing the grasshopper population from quickly reaching the comfort zone:

$$X_i^D = c\left[\sum_{j=1\, j\neq i}^{N} \frac{ub_d - lb_d}{2} * s\left(\left|x_j^D - x_i^D\right|\right) * \frac{x_j^D - x_i^D}{d_{ij}}\right] + T^D \tag{14}$$

Among, ub_d,lb_d represent the upper and lower bounds of the D-dimensional space, respectively, T^D is the value of the target in the D-dimensional. As the target wind direction, it is assumed that the target is always moving forward. C is a decreasing coefficient, which is used to gradually reduce the comfort zone, the exclusion zone, and the attraction zone until the convergence solution.

$$\mathrm{c} = c\mathrm{max} - l * \frac{\mathrm{cmax} - \mathrm{cmin}}{L} \tag{15}$$

Among, $cmax$ is the maximum value, cmin is the minimum value, l is the current iteration, L is the maximum iteration, and cmax and cmin use 1 and 0.00004.

In complicated problems with many dimensions, like planning paths for multiple robots, the usual GOA doesn't search accurately enough in the later stages and often settles too soon on a less-than-ideal solution. Scenes requiring high-precision paths and complex obstacle avoidance reflect this limitation. When multi-robots work together, maintaining the direction consistency of locust swarms during flight helps to find the global optimal solution faster and improve the efficiency of swarm behavior. Traditional GOA shows deficiencies in these aspects and cannot effectively solve the requirements of fast convergence, path smoothness, and high security in multi-robot path planning. Therefore, this paper proposes an improved grasshopper optimization algorithm (AEM-GOA).

3.2. Enhanced AEM-GOA method

To address the issues of slow convergence, path irregularities, and high collision risks in multi-robot path planning, AEM-GOA optimizes the original architecture by introducing adaptive mechanisms. Existing optimization algorithms typically employ fixed switching rules between two algorithms. This paper enhances GOA performance through three complementary adaptive mechanisms. Unlike other algorithms designed for consistent speed, it accelerates cooperative convergence via neighborhood alignment. The introduction of escape probability and random perturbation variation enhances diversity and helps the algorithm escape local optima. Collectively, these ensure the algorithm robustly handles high-dimensional constraint problems in multi-robot collaboration.

When using GOA to solve the optimal path for the robot, the individuals in the algorithm population lack directional synergy due to their randomness. This disorderly exploration method will generate many invalid searches. The slow convergence speed leads to the delay of robot decision-making, and it is difficult to respond quickly to the dynamic environment. Therefore, we introduce a velocity coordination mechanism that implements direction correction between the current solution and its neighbors, accelerating convergence through mathematical vector alignment.

$$s(r) = f_{ali} + fe^{-\frac{r}{l}} - e^{-r} \tag{16}$$

$$f_{ali} = \rho * (v_i - v_j) \tag{17}$$

Among, ρ is the direction coefficient,v_i and v_j are the velocities of two randomly selected individuals from the population, and $i \neq j$. This strategy relies solely on one alignment coefficient. This method improves the original attraction–repulsion model by making a more precise search around the best solution. The faster convergence speed enables the robot to determine its next optimal control quantity in a shorter calculation time, thereby improving the real-time and response efficiency of path planning.

In path planning, excessive emphasis on local development can easily make the algorithm fall into a suboptimal path (such as a locally unobstructed but globally distant route), while excessive emphasis on global exploration will make the algorithm difficult to converge to a sufficiently accurate solution. To overcome the exploration-utilization imbalance caused by a single update mode, a hybrid global and local exploration dynamic balance mechanism is introduced [29]. The following escape adaptive probability selection is designed. The individual chooses according to the adaptive probability. When the individual chooses to follow the local exploration that makes the execution

field interact, the individual's position is updated when the individual chooses to escape, and the jump operation is triggered to realize the global exploration:

$$X_i^D = \begin{cases} X_i^D + C_v * \left| 2r * X_{ri}^D - X_{rj}^D \right| & if \quad r_1 < r_2 \\ X_i^D * \vec{vc} & otherwise \end{cases} \tag{18}$$

Among, C_v is a linear decreasing coefficient,r、r_1、r_2 is a random number in [0,1], $\vec{vc}$ is a normal distribution vector, which is used to generate random values between. The random factor is used to select the individual update formula. When the random factor is greater than 0.5, the group force formula (14) is used to update. And dynamic exploration is used to update. The probability of switching mechanisms creates a dynamic balance between global and local search, allowing for the planning of a global path for the robot that can avoid obstacles and is generally more efficient, such as being shorter and smoother.

The solution space of the path planning problem usually has many optimal local solutions. In a dense obstacle environment, path selection may guide the robot to a dead end. It is difficult for traditional GOA to jump out of this local optimal situation. To enhance solution diversity and improve local optima avoidance, we implement a stochastic perturbation component based on adaptive stochastic perturbation distribution for the elite position, increases the random offset of zero mean and adjustable variance。

$$X_i^D = T_{best}^D + T_{gass}^D \tag{19}$$

$$T_{gass}^D = G * \left(1 - r_n^{\left(\frac{M_iter-1}{M_iter} \right)^b} \right) \tag{20}$$

$$G = \left(T_{cur}^D - T_{best}^D \right) * r_n \tag{21}$$

Among, T_{best}^D represents the best position, T_{gass}^D is the adaptive Gaussian perturbation term, G is the base perturbation vector,r_n is a random number in [1, N], b is the control parameter, which controls the update strength, T_{cur}^D is the current position. This mechanism increases the diversity of the population by generating new candidate solutions near the current optimal solution and provides an opportunity for the algorithm to escape from local extreme points. It enhances the robustness of the algorithm to find the global optimal path in a complex and constrained environment and effectively reduces the risk of the robot falling into a suboptimal or infeasible path.

Through the synergy of the above strategies, AEM-GOA systematically improves the local convergence speed, global exploration ability, and solution diversity based on maintaining the simplicity and parameter controllability of the original framework of the algorithm. As illustrated in Fig. 1, the integration of alignment control, following,

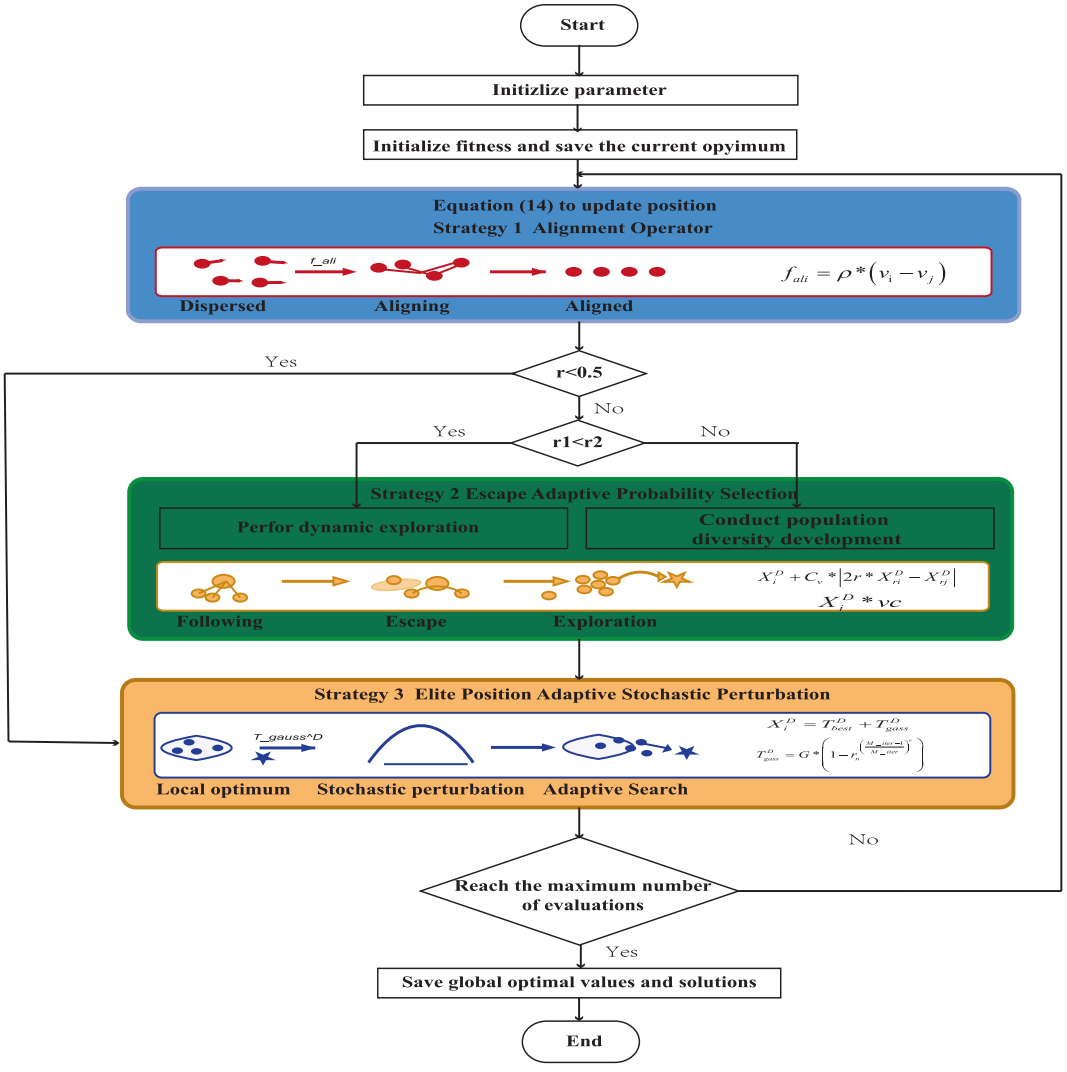


Fig. 1. Overview of the proposed AEM-GOA framework.

escaping, and adaptive stochastic perturbation enhances both the algorithm exploration and development capabilities while improving population consistency. These strategies work synergistically to ensure algorithm effectiveness and applicability in fields such as swarm robotics, path planning, and engineering optimization.

3.3. Path planning architecture based on AEM-GOA

Fig. 2 presents the comprehensive architecture of the AEM-GOA-based multi-robot path planning system, which demonstrates a systematic approach to solving complex multi-robot coordination problems. The beginnings with robot position setup, multi-constraint single-objective model construction, and initial solution population generation. In the AEM-GOA optimization mechanisms, where three enhanced strategies work synergistically: the Velocity Coordination Mechanism (Equations 16–17) ensures directional consistency among individuals, the Follow-Escape Strategy (Equations 18) dynamically balances exploration and exploitation phases, and the Random Perturbation Mutation (Equations 19–21) enhances solution diversity to escape local optima. Following position updates, the system performs dual evaluations through path quality fitness assessment and comprehensive collision detection, ensuring both optimality and safety constraints are satisfied.

The iterative nature is controlled by the termination criterion based on maximum evaluations. This framework effectively combines swarm intelligence optimization with obstacle avoidance, kinematic constraints, and real-time coordination of multi-robot systems. It clearly shows how the AEM-GOA algorithm can effectively solve the challenges of multi-robot path planning and ultimately generate smoother, safer, and more efficient paths.

4. Experiments and result analysis

The experiments were conducted using MATLAB R2024 software on a Windows 10 OSx64 operating system. The platform utilized an Intel(R) Xeon(R) CPU E5-2697 v2 @ 2.70 GHz. In this paper, the CEC2019 test suite is used to compare AEM-GOA with improved algorithms (VAGWO, WOA_LFDE, QHDBO) and state-of-the-art algorithms (GOOSE, LEA, CPO, GOA). Statistical methods including average, standard deviation, Wilcoxon rank sum test, and Friedman test are employed for evaluation.

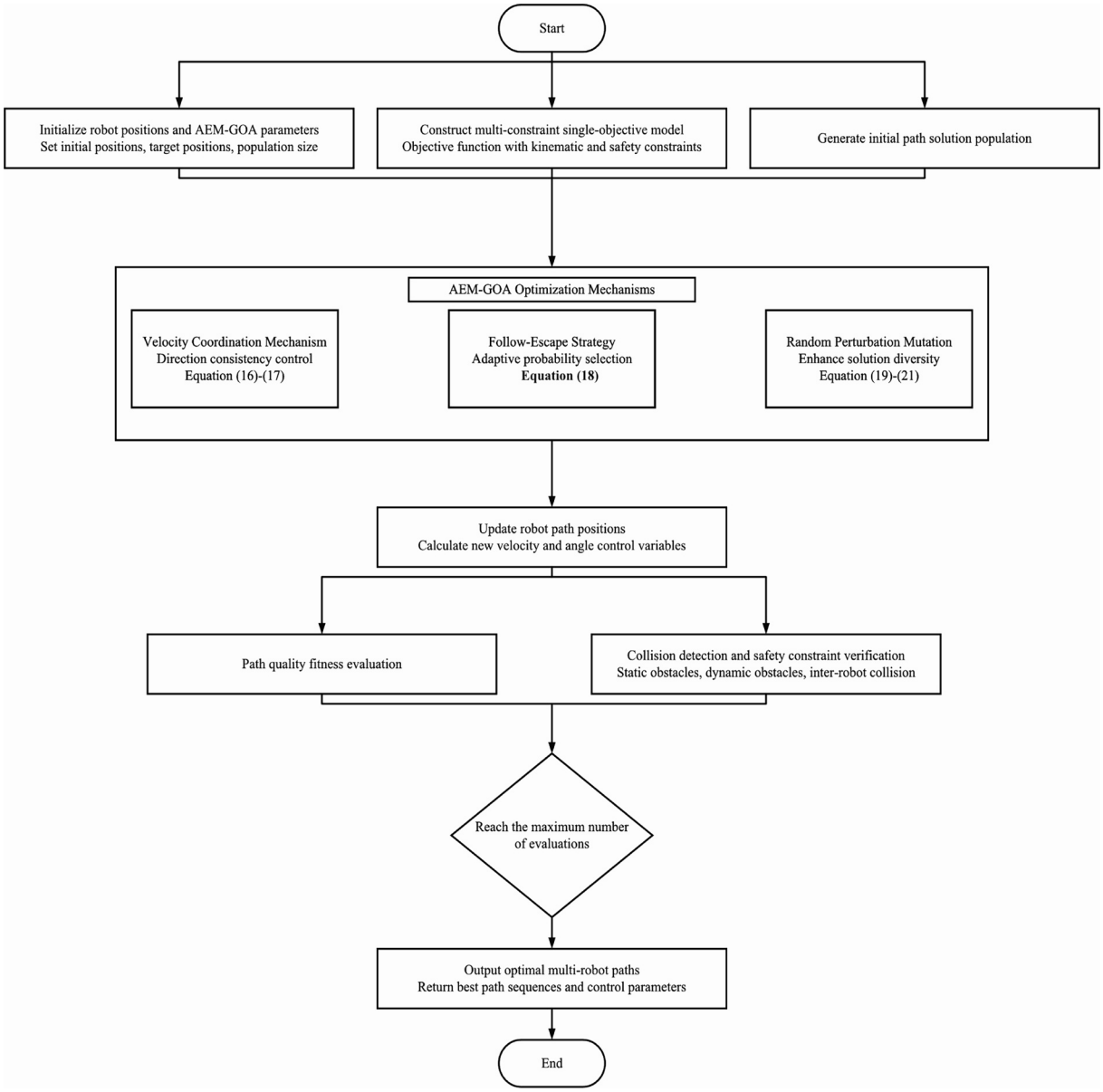


Fig. 2. Path planning flow.

The proposed method is applied to a single-objective multi-constraint path planning problem involving 16 circular robots in a 200 × 200 map with 34 static and 10 dynamic obstacles.

To ensure algorithmic robustness, AEM-GOA parameter selection is based on empirical values derived from extensive research literature on GOA, which have been validated as effective parameters for complex optimization problems [22,23]. To further validate parameter rationality, the algorithm's dynamic performance was analyzed using the CEC2019 test set.

Fig. 3 displays the search history trajectory, revealing that the population-maintained consistency without exhibiting violent oscillations, confirming the validity of the parameters. Fig. 4 illustrates the smooth transition from exploration (blue) to exploitation (red) and the specific trend of population diversity. The results indicate that the selected parameters balance the dynamic processes of global search and local optimization. This method guided by literature and validated through experiments, employs the parameter settings detailed in Table 1.

4.1. Algorithm benchmark performance validation

The comparative analysis examines the statistical performance metrics of 30 independent runs. Std is the fitness function's standard deviation of 30 independent runs. We perform the Wilcoxon rank sum test to determine if the AEM-GOA algorithm outperforms other comparable algorithms at 0.05. + means better, − means worse, and ≈ means comparable (all relative to AEM-GOA).

The CEC2019 test set consists of 10 benchmark problems, where the optimal solution is expected to be 1. The function dimensions are set to 9, 10, 16, and 18, respectively. Table 2 displays the meaning fitness value, standard, and ranking in this test set. Fig. 5 provides the Friedman ranking and Wilcoxon test results, offering further statistical validation of the algorithm's performance.

As shown in Table 3, the AEM-GOA algorithm achieves the best performance on the F1 and F3 functions with dimensions 9 and 18, attaining the lowest average fitness values. While VAGWO and WOA_LFDE exhibit comparable performance, they remain inferior to AEM-GOA. Meanwhile, GOOSE and LEA demonstrate mediocre results. Regarding solution stability, the standard deviation of AEM-GOA is closer to zero across 30 independent runs for each function, indicating high robustness due to the alignment control strategy. In contrast, other improved algorithms, such as VAGWO, WOA_LFDE, and QHDBO, exhibit poorer stability. GOOSE shows a high standard deviation of 6.84E + 08, suggesting significant instability.

In the multi-function test, AEM-GOA performs the most stably and has excellent performance on almost all test functions without obvious weaknesses. The overall performance of VAGWO was excellent, particularly in the F9 function, where it ranked first; however, it fell to seventh place in the F10 function. In contrast, the performance of WOA_LFDE fluctuates greatly. Although it wins first place in the F3 function, it is at the bottom of the F9 function. QHDBO performs best in the F2 function but ranks worst in the F3 function. The overall performance of GOOSE is weak, ranking eighth in the four functions of F4, F6, F7, and F8. Although LEA ranks first in the F5 function, it has double bottoms in the F1 and F2 functions; CPO is in the middle, but it is also at the bottom of the F10 function; although the original GOA is outstanding in the F4

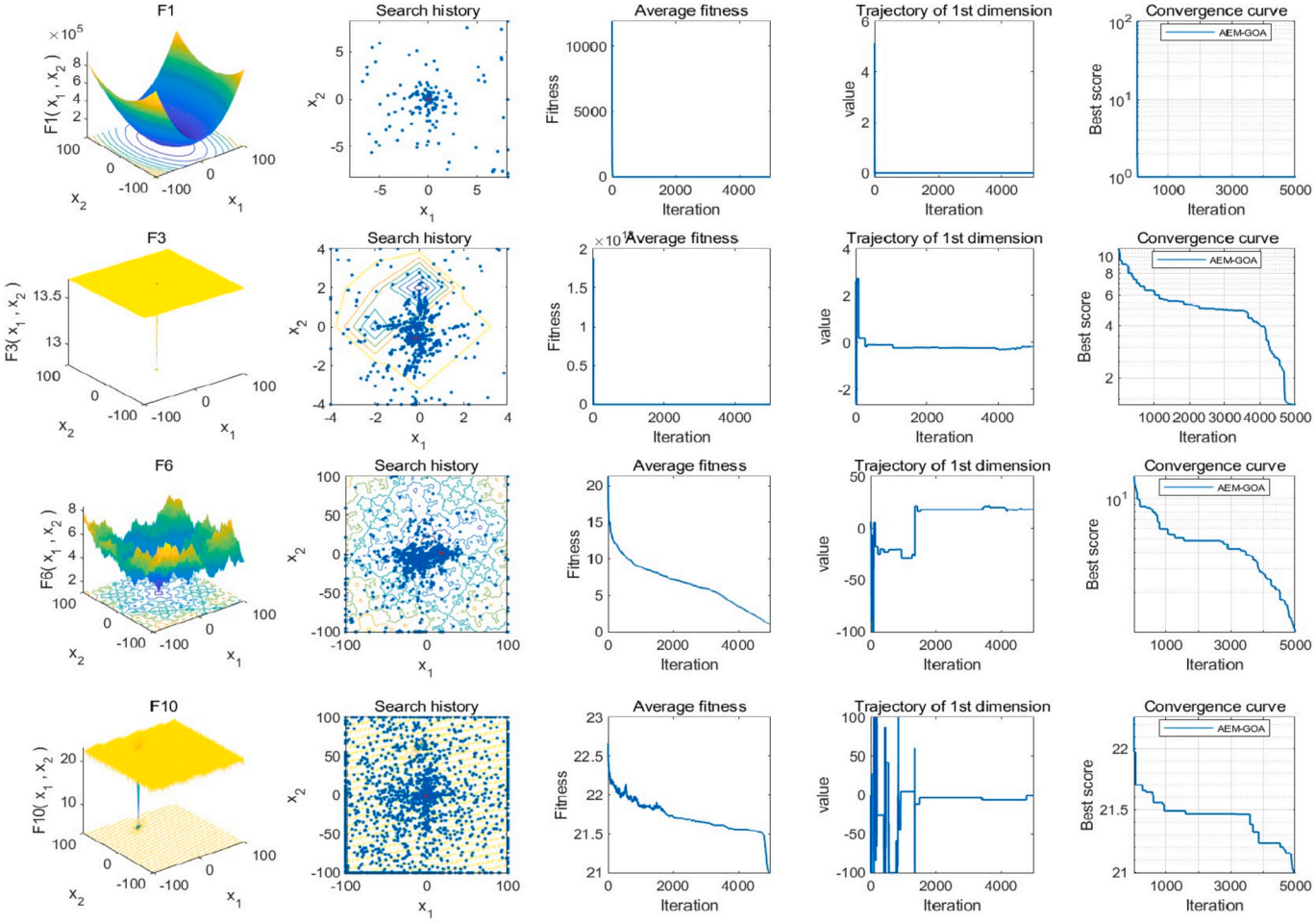


Fig. 3. Run results for CEC2019.

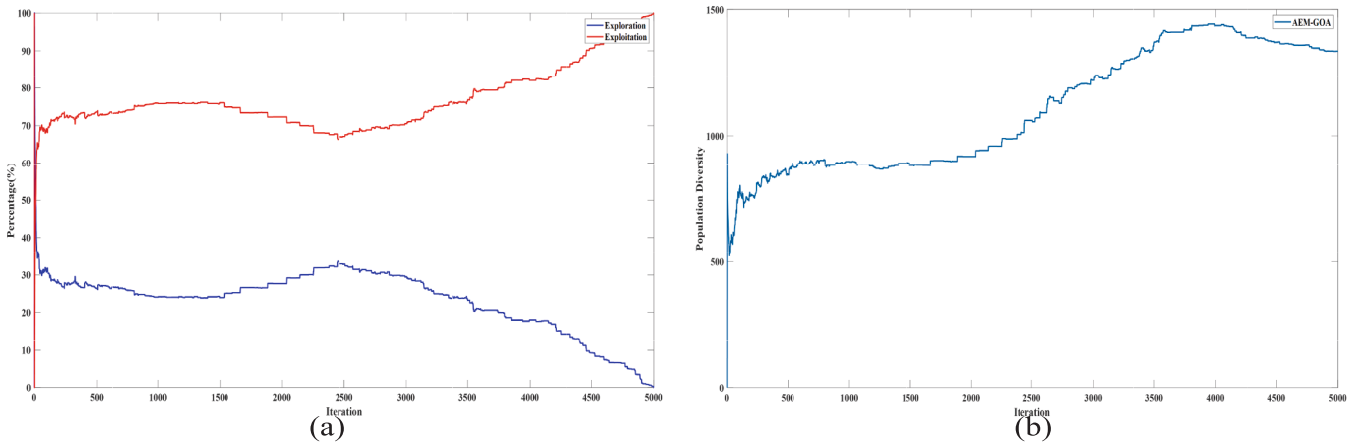


Fig. 4. Exploration, exploitation, and diversity for CEC2019.

Table 1
Parameter setting table.

Algorithm	Population size (n)	Parameters	Reference
AEM-GOA	30	cMax = 1, cMin = 0.00004, b = 1, r = rand	\
VAGWO	30	e = 1, c_max = 1, c_min = 0, k_max = 0.9, k_min = 0.4	[30]
WOA_LFDE	30	b = 1, Cr = 0.9, r = rand	[31]
QHDBO	30	P = 0.2, r = rand	[32]
GOOSE	30	coe = 0.17, r = rand[5,25]	[33]
LEA	30	h_max = 0.7, h_min = 0, lambda_c = 0.5, lambda_p = 0.5	[34]
CPO	30	N_min = 15, T = 2, alpha = 0.2, Tf = 0.8	[35]
GOA	30	cMax = 1, cMin = 0.00004, b = 1, r = rand	[21]

function (ranking first), the rest of the functions are above average.

The radar chart and line chart show that AEM-GOA has the best stability, and its performance on each test function fluctuates less. In contrast, the performance of QHDBO, WOA_LFDE, and LEA algorithms fluctuate greatly on different functions, indicating that these algorithms have significant differences in adaptability to different types of optimization problems. From the data, the Wilcoxon test result of most functions is '+', which indicates that, in comparison with the benchmark algorithm, most algorithms show statistically significant differences on most functions. This result also verifies the reliability of the performance differences between these algorithms. Fig. 6 presents the average convergence curve comparison among all algorithms.

As shown in Fig. 6, the AEM-GOA algorithm exhibits faster convergence in the F1, F3, F6, F8, and F10 test functions, enabling it to reach optimal solutions more efficiently. In F2 and F5, although its convergence speed is slightly slower than VAGWO and WOA_LFDE, it still successfully identifies the optimal solution, demonstrating superior overall convergence performance. In the comparative experiment of the CEC2019 test set, AEM-GOA demonstrates the most stable performance based on the mean optimal solution and ranking, exhibiting strong generalization and global optimization capabilities. In contrast, VAGWO and WOA_LFDE exhibit fluctuations in certain scenarios, while GOOSE and LEA show higher mean values and lower global optimization performance, making them prone to local optima.

According to the Friedman ranking and Wilcoxon test, AEM-GOA achieves the strongest overall performance across different functions, demonstrating high search efficiency, stable execution, and statistically significant advantages over other algorithms. These results confirm the effectiveness of the proposed improvement strategy, contributing to an overall performance enhancement for both improved and unimproved algorithms. From Fig. 6, in the function F1, F3, F6, F8, and F10 tests, the AEM-GOA algorithm converges faster and can determine most solutions faster. In the functions F2 and F5, the convergence speed is less than VAGWO and WOA_LFDE, but the optimal value can still be found, and the overall convergence performance is better.

4.2. *Multi-robot path planning simulation and analysis*

To evaluate the performance of the proposed framework in solving distributed cooperation problems in dynamic environments, we selected state-of-the-art swarm intelligence algorithms for comparison. Specifically, the proposed AEM-GOA is compared with SDSCA [28]and VAGWO [29], which are representative advanced *meta*-heuristic methods for multi-robot path planning. Traditional graph-search or sampling-based planners were not included in this comparison as the focus of this study is on verifying the specific improvements in convergence speed and obstacle avoidance within the swarm intelligence domain. Each algorithm runs 5 times, each iteration 200 times, and the population is set to 30. The study compares the average motion deviation, average running step size, average cost function value, and collision rate.

Each robot designates a target at varying distances during the path planning evaluation, and it executes five distinct maps featuring diverse obstacles. Fig. 7 illustrates the schematic of three algorithm-based path planning methods across five scenarios. Table 4 presents the average collision rate, success rate, and average path deviation error (APDE) for the five runs. The *meta*-heuristic algorithm has created a workable path, showing that the multi-constraint single-objective model from the previous section can be used. Furthermore, the algorithm has identified the optimal solution within the model, indicating its proficient capability in addressing the path planning model.

Fig. 7 path results and Table 4 data show that AEM-GOA has no collision in all five runs, the collision rate is 0 %, the success rate is 100 %, the average path deviation (APDE) is only 80.94, and the average running time is 3549.6 s. Although SDSCA shows excellent time efficiency with the shortest time of 3263.9 s, it exposes its risk in complex environments with a maximum collision rate of 1.92 % and a maximum deviation of 339.85. The collision rate (1.64 %), success rate (98.36 %), and APDE (155.83) of VAGWO are at a medium level, but its average time of 5286.5 s is significantly longer than the other two.

Generally, AEM-GOA maintains a large distance from the obstacle, and the generated trajectory is the smoothest. VAGWO only has 8 slight collisions, and the path is coherent. Although SDSCA excels in path length and search speed, its excessive pursuit of efficiency results in multiple trajectories that are close to obstacles, significantly increasing

Table 2
Average and Std results of CEC2019 Test Suite.

	Index	AEM-GOA	VAGWO	WOA_LFDE	QHDBO	GOOSE	LEA	CPO	GOA
F1	Avg	**1.00E + 00**	1.96E + 04	4.30E + 03	1.96E + 02	3.11E + 08	7.43E + 06	1.00E + 00	3.84E + 05
	Std	0.00E + 00	2.72E + 04	1.49E + 04	1.04E + 03	6.84E + 08	4.51E + 06	0.00E + 00	2.58E + 05
	Best	1.00E + 00	1.76E + 01	1.00E + 00	1.00E + 00	1.00E + 00	5.25E + 05	1.00E + 00	2.63E + 04
	Worst	1.00E + 00	5.71E + 03	1.16E + 02	1.00E + 00	4.25E + 01	6.81E + 06	1.00E + 00	3.44E + 05
	Avg.rank	**1.00E + 00**	5.00E + 00	4.00E + 00	3.00E + 00	8.00E + 00	7.00E + 00	1.00E + 00	6.00E + 00
	Std.rank	**1.00E + 00**	5.00E + 00	4.00E + 00	3.00E + 00	8.00E + 00	7.00E + 00	1.00E + 00	6.00E + 00
F2	Avg	4.48E + 00	3.40E + 02	6.63E + 02	**4.28E + 00**	1.46E + 04	2.85E + 03	4.84E + 00	3.69E + 02
	Std	2.70E-01	1.31E + 02	7.92E + 02	1.40E-01	1.48E + 04	1.80E + 03	2.00E-01	1.40E + 02
	Best	4.28E + 00	6.63E + 01	1.47E + 02	4.11E + 00	4.18E + 00	5.44E + 02	4.37E + 00	1.89E + 02
	Worst	4.36E + 00	3.19E + 02	4.14E + 02	4.26E + 00	1.35E + 04	2.38E + 03	4.97E + 00	3.46E + 02
	Avg.rank	2.00E + 00	4.00E + 00	6.00E + 00	**1.00E + 00**	8.00E + 00	7.00E + 00	3.00E + 00	5.00E + 00
	Std.rank	3.00E + 00	4.00E + 00	6.00E + 00	**1.00E + 00**	8.00E + 00	7.00E + 00	2.00E + 00	5.00E + 00
F3	Avg	**1.33E + 00**	3.08E + 00	1.75E + 00	1.37E + 01	8.13E + 00	7.68E + 00	6.98E + 00	7.51E + 00
	Std	2.10E-01	2.21E + 00	9.30E-01	0.00E + 00	3.07E + 00	1.81E + 00	9.20E-01	2.37E + 00
	Best	1.00E + 00	1.00E + 00	1.41E + 00	1.37E + 01	1.41E + 00	3.28E + 00	5.23E + 00	1.41E + 00
	Worst	1.41E + 00	1.41E + 00	1.41E + 00	1.37E + 01	9.71E + 00	7.70E + 00	6.90E + 00	7.71E + 00
	Avg.rank	**1.00E + 00**	3.00E + 00	2.00E + 00	8.00E + 00	7.00E + 00	6.00E + 00	4.00E + 00	5.00E + 00
	Std.rank	2.00E + 00	6.00E + 00	4.00E + 00	**1.00E + 00**	8.00E + 00	5.00E + 00	3.00E + 00	7.00E + 00
F4	Avg	2.35E + 01	2.19E + 01	3.96E + 01	2.74E + 01	5.07E + 01	3.90E + 01	3.73E + 01	**1.80E + 01**
	Std	5.76E + 00	9.10E + 00	1.42E + 01	7.79E + 00	1.80E + 01	1.28E + 01	5.92E + 00	8.58E + 00
	Best	1.39E + 01	5.97E + 00	1.79E + 01	1.39E + 01	2.99E + 01	1.79E + 01	2.38E + 01	6.97E + 00
	Worst	2.19E + 01	2.45E + 01	3.70E + 01	2.67E + 01	4.33E + 01	3.55E + 01	3.74E + 01	1.59E + 01
	Avg.rank	3.00E + 00	2.00E + 00	7.00E + 00	4.00E + 00	8.00E + 00	6.00E + 00	5.00E + 00	**1.00E + 00**
	Std.rank	**1.00E + 00**	5.00E + 00	7.00E + 00	3.00E + 00	8.00E + 00	6.00E + 00	2.00E + 00	4.00E + 00
F5	Avg	1.25E + 00	1.41E + 00	2.16E + 00	1.26E + 00	1.84E + 00	**1.14E + 00**	1.62E + 00	1.33E + 00
	Std	9.00E-02	2.20E-01	1.13E + 00	2.50E-01	1.24E + 00	9.00E-02	1.50E-01	1.30E-01
	Best	1.11E + 00	1.06E + 00	1.09E + 00	1.05E + 00	1.09E + 00	1.02E + 00	1.34E + 00	1.15E + 00
	Worst	1.23E + 00	1.49E + 00	1.75E + 00	1.16E + 00	1.46E + 00	1.12E + 00	1.66E + 00	1.32E + 00
	Avg.rank	2.00E + 00	5.00E + 00	8.00E + 00	3.00E + 00	7.00E + 00	**1.00E + 00**	6.00E + 00	4.00E + 00
	Std.rank	**1.00E + 00**	5.00E + 00	7.00E + 00	6.00E + 00	8.00E + 00	1.00E + 00	4.00E + 00	3.00E + 00
F6	Avg	**1.21E + 00**	1.69E + 00	6.78E + 00	4.79E + 00	1.23E + 01	5.51E + 00	2.92E + 00	2.90E + 00
	Std	4.70E-01	7.50E-01	1.54E + 00	1.31E + 00	1.52E + 00	1.50E + 00	8.30E-01	1.22E + 00
	Best	1.00E + 00	1.02E + 00	3.22E + 00	1.62E + 00	9.44E + 00	3.02E + 00	1.81E + 00	1.01E + 00
	Worst	1.04E + 00	1.25E + 00	6.32E + 00	5.03E + 00	1.24E + 01	5.60E + 00	2.58E + 00	2.60E + 00
	Avg.rank	**1.00E + 00**	2.00E + 00	7.00E + 00	5.00E + 00	8.00E + 00	6.00E + 00	4.00E + 00	3.00E + 00
	Std.rank	**1.00E + 00**	2.00E + 00	8.00E + 00	5.00E + 00	7.00E + 00	6.00E + 00	3.00E + 00	4.00E + 00
F7	Avg	2.13E + 02	**2.09E + 02**	3.70E + 02	6.99E + 02	1.16E + 03	4.50E + 02	7.14E + 02	7.34E + 02
	Std	1.06E + 02	1.38E + 02	2.38E + 02	2.94E + 02	3.53E + 02	2.25E + 02	1.67E + 02	2.99E + 02
	Best	1.62E + 01	4.74E + 00	2.62E + 01	2.45E + 02	4.55E + 02	3.13E + 01	2.84E + 02	2.62E + 02
	Worst	1.38E + 02	1.61E + 02	3.10E + 02	7.06E + 02	1.21E + 03	3.94E + 02	7.09E + 02	7.35E + 02
	Avg.rank	2.00E + 00	**1.00E + 00**	3.00E + 00	5.00E + 00	8.00E + 00	4.00E + 00	6.00E + 00	7.00E + 00
	Std.rank	**1.00E + 00**	2.00E + 00	5.00E + 00	6.00E + 00	8.00E + 00	4.00E + 00	3.00E + 00	7.00E + 00
F8	Avg	**2.97E + 00**	3.42E + 00	4.17E + 00	3.81E + 00	5.36E + 00	4.39E + 00	4.33E + 00	3.81E + 00
	Std	6.30E-01	6.40E-01	4.40E-01	5.90E-01	2.60E-01	2.80E-01	1.50E-01	5.60E-01
	Best	1.72E + 00	2.15E + 00	3.04E + 00	2.54E + 00	4.70E + 00	3.80E + 00	3.97E + 00	2.84E + 00
	Worst	3.08E + 00	3.43E + 00	4.18E + 00	3.99E + 00	5.44E + 00	4.36E + 00	4.34E + 00	3.99E + 00
	Avg.rank	**1.00E + 00**	2.00E + 00	5.00E + 00	3.00E + 00	8.00E + 00	7.00E + 00	6.00E + 00	3.00E + 00
	Std.rank	7.00E + 00	8.00E + 00	4.00E + 00	6.00E + 00	2.00E + 00	3.00E + 00	**1.00E + 00**	5.00E + 00
F9	Avg	1.11E + 00	**1.08E + 00**	1.55E + 00	1.49E + 00	1.49E + 00	1.49E + 00	1.42E + 00	1.13E + 00
	Std	3.00E-02	4.00E-02	1.50E-01	1.70E-01	2.60E-01	2.40E-01	8.00E-02	6.00E-02
	Best	1.04E + 00	1.01E + 00	1.24E + 00	1.16E + 00	1.19E + 00	1.17E + 00	1.26E + 00	1.05E + 00
	Worst	1.11E + 00	1.07E + 00	1.54E + 00	1.49E + 00	1.45E + 00	1.46E + 00	1.42E + 00	1.11E + 00
	Avg.rank	2.00E + 00	**1.00E + 00**	8.00E + 00	5.00E + 00	5.00E + 00	5.00E + 00	4.00E + 00	3.00E + 00
	Std.rank	**1.00E + 00**	2.00E + 00	5.00E + 00	6.00E + 00	8.00E + 00	7.00E + 00	4.00E + 00	3.00E + 00
F10	Avg	**1.97E + 01**	2.07E + 01	2.10E + 01	2.13E + 01	2.10E + 01	2.11E + 01	2.15E + 01	2.11E + 01
	Std	5.07E + 00	3.71E + 00	7.00E-02	1.30E-01	0.00E + 00	1.40E-01	2.20E-01	7.00E-02
	Best	1.00E + 00	1.00E + 00	2.10E + 01	2.10E + 01	2.10E + 01	2.10E + 01	2.04E + 01	2.10E + 01
	Worst	2.10E + 01	2.13E + 01	2.10E + 01	2.13E + 01	2.10E + 01	2.11E + 01	2.15E + 01	2.10E + 01
	Avg.rank	**1.00E + 00**	2.00E + 00	4.00E + 00	7.00E + 00	3.00E + 00	6.00E + 00	8.00E + 00	5.00E + 00
	Std.rank	8.00E + 00	7.00E + 00	2.00E + 00	4.00E + 00	**1.00E + 00**	5.00E + 00	6.00E + 00	2.00E + 00
	Avg.rank.v	1.60E + 00	2.70E + 00	5.40E + 00	4.40E + 00	7.00E + 00	5.50E + 00	4.70E + 00	4.20E + 00
	Com.rank	1.00E + 00	2.00E + 00	6.00E + 00	4.00E + 00	8.00E + 00	7.00E + 00	5.00E + 00	3.00E + 00
	Std.rank.v	2.60E + 00	4.60E + 00	5.20E + 00	4.10E + 00	6.60E + 00	5.10E + 00	2.90E + 00	4.60E + 00
	Com.rank	1.00E + 00	4.00E + 00	7.00E + 00	3.00E + 00	8.00E + 00	6.00E + 00	2.00E + 00	4.00E + 00

the risk of collision. APDE, as an index to measure the degree between the algorithm planning path and the ideal shortest path, also reflects the differences in the optimization ability of the three. The deviation value of AEM-GOA is 48 % and 76 % lower than that of VAGWO and SDSCA, respectively, indicating that it has the strongest approximation ability to the global optimal trajectory, followed by VAGWO, while the high deviation of SDSCA reflects its sacrifice of path smoothness and global optimization degree.

Fig. 8 shows the function optimization of the three functions in five scenarios, and the function values are minimized. Fig. 9 presents the average step length and standard deviation of the robot across five scenarios, illustrating the algorithm's efficiency and assessing its ability to maintain this efficiency in various situations, thereby ensuring the objectivity and reliability of the evaluation.

From the perspective of convergence and stability, the iterated function Fig. 8 shows that AEM-GOA not only converges fast but also has

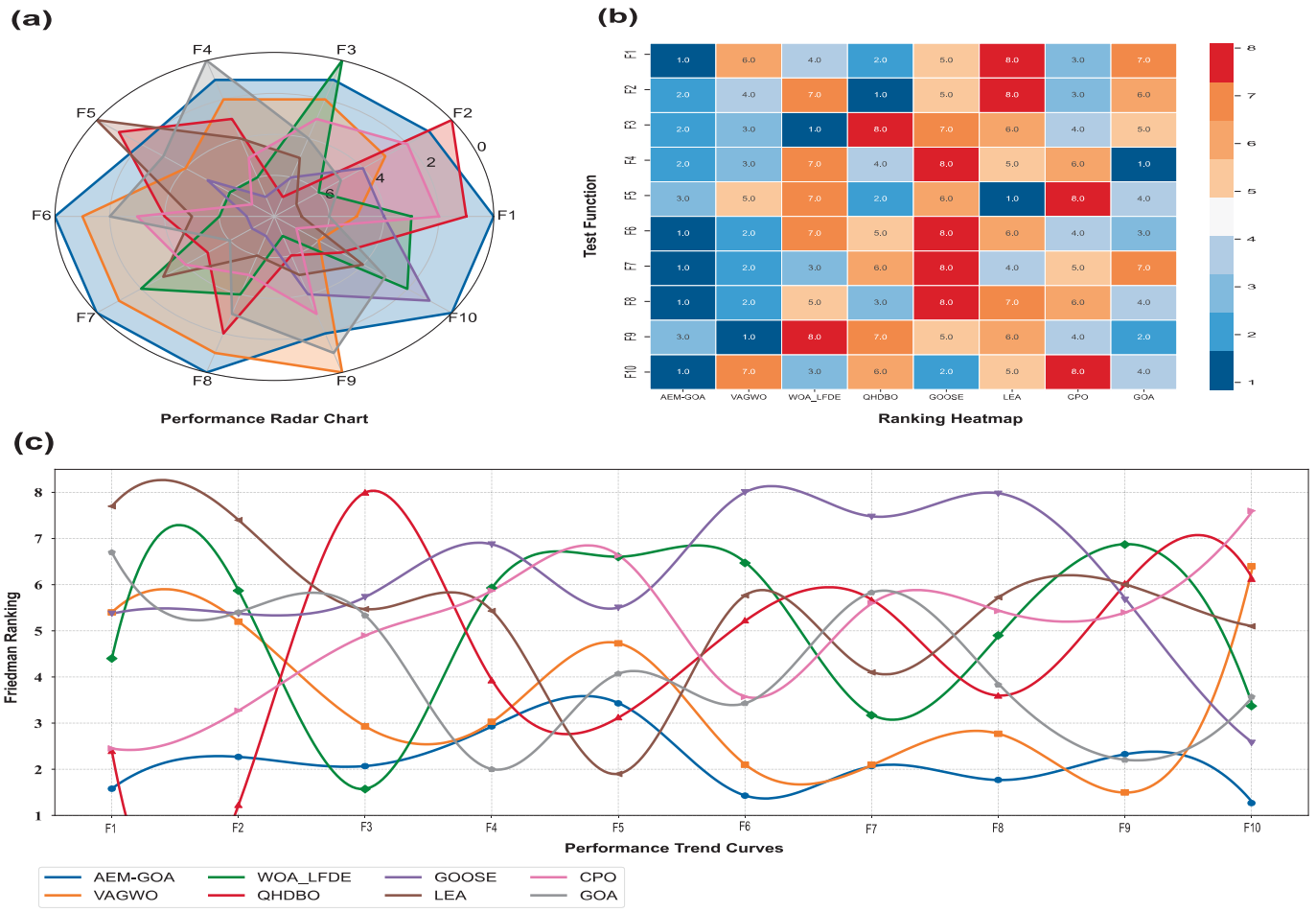


Fig. 5. Algorithm flow figure for CEC2019.

the smoothest curve and finally obtains the lowest objective function value. VAGWO can also reach a similar optimal level in several scenarios, but the overall fluctuation is large. Although SDSCA tends to stabilize rapidly, it stays at a relatively high function value. We conducted a further analysis of the step size. Fig. 9 shows that AEM-GOA maintains the optimal average step size and the minimum standard deviation in most scenarios, indicating that it can maintain consistent efficiency in different test environments. While VAGWO performs well, the increase in target point distance significantly elongates SDSCA, leading to significant fluctuations in efficiency.

The AEM-GOA algorithm maintains high real-time performance and stability while ensuring zero collision and low path deviation, demonstrating optimal comprehensive performance and robustness. VAGWO performs well in accuracy and convergence quality, but the running time is long. SDSCA is more suitable for task scenarios with time efficiency requirements and a relatively single environment. Each of the three algorithms has its emphasis, which provides an alternative and high-performance solution for multi-constrained single-objective multi-robot path planning.

5. Discussion

The experimental results show that the proposed AEM-GOA has excellent performance in path quality, safety, and smoothness in multi-robot path planning tasks. Compared with SDSCA and VAGWO algorithms, AEM-GOA maintains higher robustness in multiple verifications. The performance advantage of the algorithm is verified in the CEC2019 benchmark test, which is better than the benchmark GOA and other seven algorithms, indicating that AEM-GOA has strong adaptability in complex constraint optimization tasks.

A key finding from the path simulations is the superior smoothness of trajectories generated by AEM-GOA, which exhibit significantly less unnecessary steering and jitter. The novel follows the escape adaptive probability strategy directly explains this phenomenon. Unlike traditional algorithms that may over-exploit a local region, leading to sharp turns, our strategy dynamically balances exploitation and exploration. By adaptively increasing the probability of 'escaping' when the population stagnates, the algorithm is prompted to seek broader, more globally efficient routes, naturally avoiding the jagged paths characteristic of premature convergence in cluttered environments. The strategy suppresses the risk of premature convergence in the local search process by dynamically adjusting the search behavior probability, thereby maintaining the global search ability in a complex environment and reducing the probability of the path falling into the local optimum. In addition, AEM-GOA can continuously obtain high-quality solutions in multiple independent operations, mainly due to the synergy between the speed coordination mechanism and the disturbance variation mechanism.

Beyond numerical metrics, the results hold physical significance for real-world coordination. The convergence speed directly translates to real-time responsiveness, reducing computational latency in dynamic obstacle avoidance. Path smoothness reflects control stability, with AEM-GOA generating smooth trajectories that minimize motion jitter. This directly reduces mechanical wear on actuators in engineering practice. These advantages stem from the follow-escape, adaptive probability, and velocity coordination mechanisms, which dynamically balance exploration and exploitation to ensure globally efficient and safe path planning.

Table 3
Friedman and Wilcoxon results of CEC2019 Test Suite.

	Index	AEM-GOA	VAGWO	WOA_LFDE	QHDBO	GOOSE	LEA	CPO	GOA
F1	Friedman	1.58	5.40	4.40	2.40	5.37	7.70	2.45	6.70
	Wilcoxon	\	+	+	+	+	+	+	+
	Fri.rank	**1.00**	6.00	4.00	2.00	5.00	8.00	3.00	7.00
F2	Friedman	2.27	5.20	5.87	1.23	5.37	7.40	3.27	5.40
	Wilcoxon	\	+	+	−	+	+	+	+
	Fri.rank	2.00	4.00	7.00	**1.00**	5.00	8.00	3.00	6.00
F3	Friedman	2.07	2.93	1.57	8.00	5.73	5.47	4.90	5.33
	Wilcoxon	\	≈	≈	+	+	+	+	+
	Fri.rank	2.00	3.00	**1.00**	8.00	7.00	6.00	4.00	5.00
F4	Friedman	2.93	3.03	5.93	3.93	6.87	5.43	5.87	2.00
	Wilcoxon	\	≈	+	≈	+	+	+	−
	Fri.rank	2.00	3.00	7.00	4.00	8.00	5.00	6.00	**1.00**
F5	Friedman	3.43	4.73	6.60	3.13	5.50	1.90	6.63	4.07
	Wilcoxon	\	+	+	≈	+	−	+	+
	Fri.rank	3.00	5.00	7.00	2.00	6.00	**1.00**	8.00	4.00
F6	Friedman	1.43	2.10	6.47	5.23	8.00	5.77	3.57	3.43
	Wilcoxon	\	+	+	+	+	+	+	+
	Fri.rank	**1.00**	2.00	7.00	5.00	8.00	6.00	4.00	3.00
F7	Friedman	2.07	2.10	3.17	5.67	7.47	4.10	5.60	5.83
	Wilcoxon	\	≈	+	+	+	+	+	+
	Fri.rank	**1.00**	2.00	3.00	6.00	8.00	4.00	5.00	7.00
F8	Friedman	1.77	2.77	4.90	3.60	7.97	5.73	5.43	3.83
	Wilcoxon	\	+	+	+	+	+	+	+
	Fri.rank	**1.00**	2.00	5.00	3.00	8.00	7.00	6.00	4.00
F9	Friedman	2.33	1.50	6.87	6.03	5.67	6.00	5.40	2.20
	Wilcoxon	\	−	+	+	+	+	+	≈
	Fri.rank	3.00	**1.00**	8.00	7.00	5.00	6.00	4.00	2.00
F10	Friedman	1.27	6.40	3.37	6.13	2.57	5.10	7.60	3.57
	Wilcoxon	\	+	+	+	+	+	+	+
	Fri.rank	**1.00**	7.00	3.00	6.00	2.00	5.00	8.00	4.00
	Avg.rank	1.70	3.50	5.20	4.40	6.20	5.60	5.10	4.30
	Com.rank	**1.00**	2.00	6.00	4.00	8.00	7.00	5.00	3.00

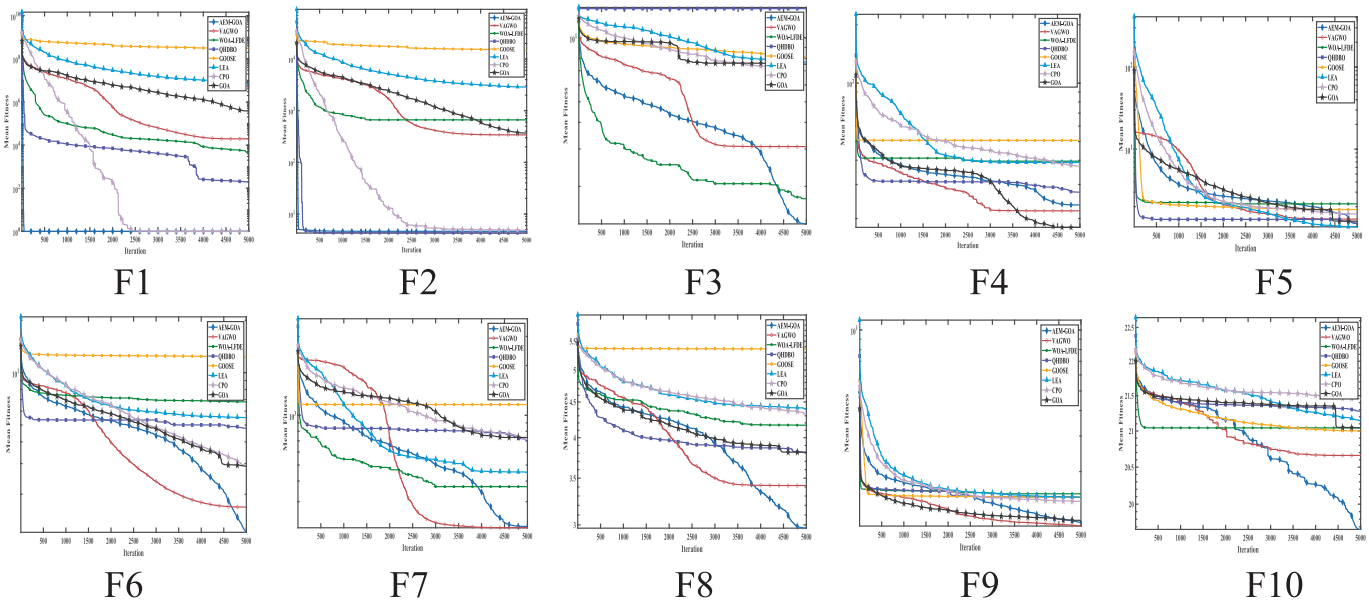


Fig. 6. Convergence corves on CEC2019.

6. Conclusion

This study proposes the AEM-GOA algorithm. By introducing the speed coordination mechanism, the follow-escape adaptive probability strategy, and the adaptive random disturbance mutation, the convergence performance and optimization ability of the algorithm are improved. Compared to the traditional GOA, AEM-GOA significantly enhances both convergence speed and optimization accuracy; additionally, it is a simple algorithm with few parameters, achieving a satisfactory balance between exploration and development. The experimental results show that the improved algorithm shows stronger robustness and performance advantages in multiple benchmarks. In the application of multi-robot path planning, the AEM-GO algorithm achieves a 100 % success rate, the average path deviation error is only 80.94, and the optimal fitness function value is obtained in the optimization process. The algorithm generates the smoothest trajectory while ensuring security, showing the strongest global trajectory optimization ability.

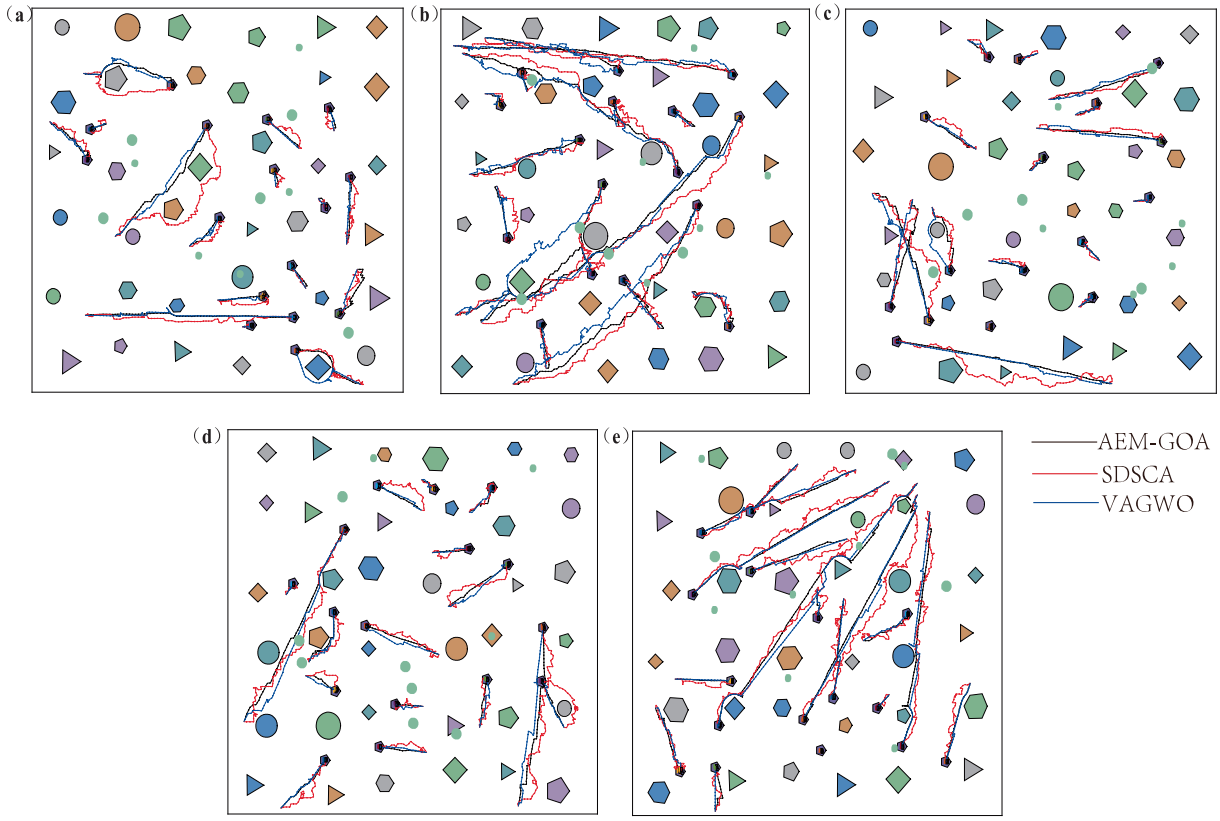


Fig. 7. The planning paths generated by three comparison algorithms.

Table 4
Comparison of path planning performance for each algorithm.

Algorithm	AEM-GOA	SDSCA	VAGWO
Collision	0	11	8
Normals	467	561	481
Totals	467	572	489
Collision Rate (%)	0	1.92	1.64
Success Rate (%)	100	98.08	98.36
APDE	80.94	339.85	155.83
Time(s)	3549.6	3263.87	5286.5

However, the integration of multiple improvement mechanisms inevitably introduces higher computational complexity compared to basic algorithms. Therefore, future work will focus on the following aspects:

1 Focusing on reducing the computational dimension of the algorithm to improve its real-time adaptability in practical engineering applications of multi-robot path planning.
2 Planning to investigate the scalability of the proposed framework for larger robot teams to ensure that the adaptive coordination mechanisms remain robust as the number of agents increases.
3 Validating the algorithm on physical mobile robot platforms to verify its effectiveness in real-world scenarios such as intelligent warehousing and disaster rescue operations.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Zheng Liu: Writing – original draft, Software, Methodology, Data

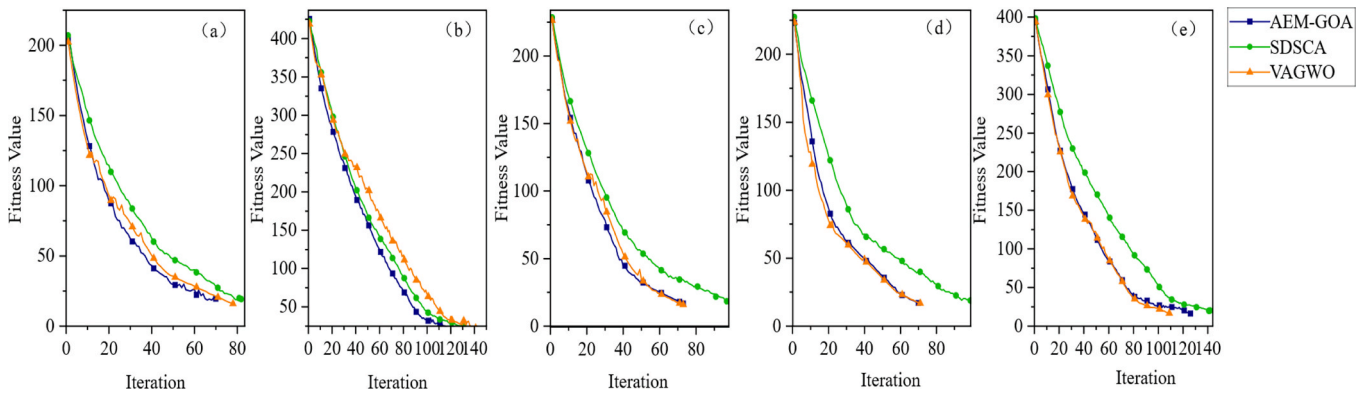


Fig. 8. Iteration function curve.

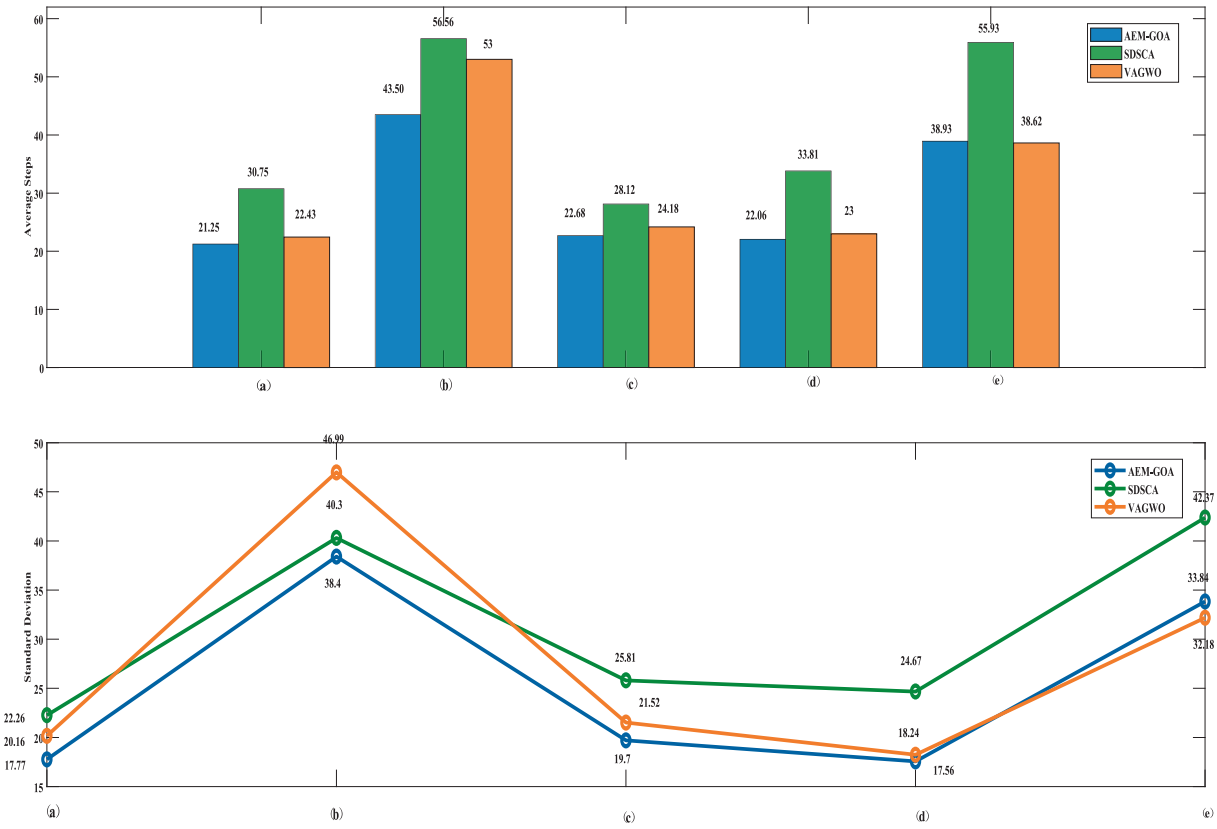


Fig. 9. Average step and std.

curation. **Wan Xu:** Writing – review & editing, Methodology, Conceptualization. **Yujie Wang:** Visualization, Software, Data curation. **Xiaochi Mu:** Visualization, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

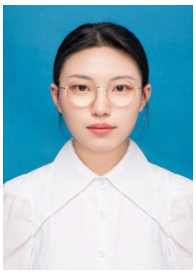
Acknowledgments

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